

The Nehalem Cometh

The fastest CPU on the planet, the QX 9770, is in severe danger. It faces competition; but not from the expected quarter i.e. AMD. Shockingly, Intel is doing to itself what they did to AMD two years ago with their Core 2 Duos. They promised a new CPU architecture every two years and true to their promise we got a large, inconspicuous looking parcel from Intel, USA. Opening it we were surprised to find a large black motherboard called the DX 58SO, nicknamed "SmackOver"; obviously based on their X58 chipset. Also in the bundle was a rather large CPU, clearly larger than any LGA 775 CPU. We also found a set of three 1-GB DDR3 1066 sticks. Now Nehalem supports tri-channel DDR3, so by now we were rubbing our hands in glee. Rummaging deep into the carton we unpacked a small box that contained an 80 GB Solid State Drive that runs off a SATA 2.0 interface.

Setting up the kit, we had a quick look at the DX 5850s BIOS which was laden with over-clocking features and all sorts of settings; enough to keep even hardcore overclockers exuberant. Interestingly the Nehalem runs off a 133 MHz FSB, while all Penryn CPUs use a faster 333 MHz bus. Thus the multiplier for these CPUs is very high. We found



The Nehalem Kit consisting of X58 motherboard, 3 GB of 1066 MHz DDR3 RAM and Intel 80GB SSD

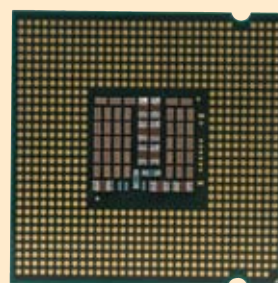


we had the Core i7 Extreme 965, which is a 3.2 GHz part and the top-of-the-line Nehalem processor — oh joy! There was no way to set the multiplier to run at 24x fixed. An option to turn on *Turbo Mode* promised to freeze the multiplier to any maximum value specified (in this case 24x) but this didn't work and the multiplier shuffled between 12x and 24x throughout our tests.

The reason for Nehalem having a slower bus is that the CPU

itself now has the memory controller integrated on its die; a first for Intel. Therefore a very fast bus is no longer needed to communicate with the Northbridge, which traditionally incorporated the memory controller for Intel-based systems. Nehalem should now (at least theoretically) enjoy very low latency access to memory resulting in improved memory utilisation, something which AMD has excelled at since their memory controllers became part of their CPU dies. This in itself is the biggest change and something that makes Nehalem totally different from the Core 2 architecture and

	i7 Extreme 965 (3.2 GHz)	QX 9770 (3.2 GHz)	Difference (In Percent)
PC Mark05 (CPU Score)	10897	10321	106
3D Mark 2006 (CPU Score)	6411	5238	122
Wprime (32 Million) (all cores / single core)	8.328	13.296	160
ScienceMark 2.0			
Cryptography (AES)	8.503	8.933	105
Membench	13165.14	7699.68	171
CineBench R10			
CPU Rendering (All CPUs)	15609	12579	124
CPU Rendering (1 CPU)	3845	3522	109
PovRay 3.7 (Real Time)			
Multi CPU Bench	91.17	87.63	96
Single CPU Bench	312.88	343.78	110
Chess 2			
Maya 8.0 Rendering (sec, all cores)	19.9	20.9	105
Maya 8.0 Rendering (sec, single core)	43.5	52.3	120
WinRAR 3.8 (KBps)	4109	2156	191
DivX Encode 6.8.0.19 (sec)	91.5	107.1	117
UT 3 (Deimos)	62	216	29
Doom 3	378.5	311.3	122
Company Of Heroes	606.6	570.2	106

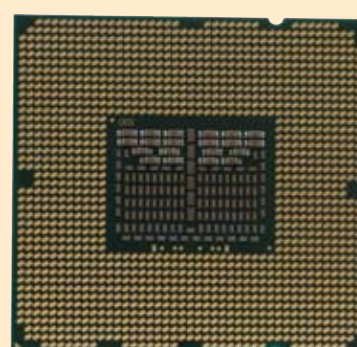


Intel QX9770 on the left, Intel i7 Extreme 965 on the right. Note the increase in die size

something that was missing from Intel's repertoire for awhile now. Owing to this shift, all Nehalem CPUs are built around a colossal 1366 LGA socket, a huge hike in number of contacts from LGA 775. Nehalem also features an old trick from Intel - Hyper-threading; which hasn't been around since the third generation of Pentium 4 processor. Nehalem is available as dual- and quad-core configurations, which means four or eight hyper threaded cores. True to this Windows XP shows the i7 Extreme 965 as having eight cores.

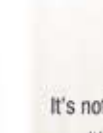
Besides this Nehalem uses L3 cache, previously unheard of on desktop CPUs except for Intel's Extreme Edition Pentium 4s of old. This L3 cache is shared between cores, much like Core 3 Duo processors share their L2 cache. The L2 cache on Nehalem CPUs is now discrete per core; and each core gets 256 KB of dedicated L2 cache. The L3 cache is a whopping 8 MB, though still not as much as the 12 MB L2 caches on the Yorkfield processors. Nehalem is still not natively quad core. Due to the fact that the i7 has an integrated memory controller its external bus operates on a HT-like link at 6.4 GBps. This in itself means the numbers are incomparable to earlier Intel CPUs including the Wolfdale and Yorkfield processors.

We tested the Nehalem kit with the supplied DX 58SO motherboard, 1 x 3 GB DIMMs of DDR3 1066 MHz memory, Intel's 80 GB SSD and ASUS' GTX 280. Although a fair comparison isn't possible since we used a number of different components on our CPU test bench we did a few comparisons. The memory on the Nehalem system was running at 1066 MHz at



timings of 7-7-7-20, which is possibly slower than DDR2 800 MHz at 4-4-4-12; which means the QX 9770 has somewhat of an unfair advantage. Of course with read speeds in the region of 260 MB/s, SSD is a solid advantage for the newcomer, so we're calling it about even. With identically clocked core speeds both these CPUs slugged it out, but Nehalem seems to have a trick too many for the QX 9770 which gives up in despair. Only in UT 3 did we see a curious 62 fps drop- this has got to be a glitch with the game on this platform and updates from either Intel or Epic Games are sure to remedy this. The only other benchmark to favour the older QX 9770 is the multi-core benchmark in POVRay 3.7, where the i7 965 is some 3.5 seconds slower in completing the scene, this may be attributed to the overhead involved in hyper threading, because in the single core render test Nehalem is faster.

What was shocking was the difference HT made under wPrime; it's a whole 60 per cent faster; Nehalem chalks up 8.503 seconds, something a Core 2 Quad wouldn't do even at 4 GHz, and something most dual cores cannot dream of even at beyond 5 GHz! Likewise all applications show a healthy performance gain, WinRAR being the biggest beneficiary to the newcomers ego with a super 91 per cent increment in compression/decompression performance. The best thing is that this is a review kit, with initial BIOS updates and drivers; as the platform matures the i7 will get a lot faster. Faster DDR3 memory will also help. If that wasn't enough the top-end i7 Extreme 965 is expected to cost around Rs 55,000, which makes it cheaper than the QX 9770.



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